

Vaidhyaa Premium Application Design Framework

A functional source of truth for redesigning the clinic portal, doctor workspace, AI receptionist operations and internal platform tools

This document defines what the application does, who may perform each action, which behavior is live or partial, and how the frontend should be redesigned without changing business rules. It is intended for a design-generation system, product designer or frontend implementation team.

Document	Value
Version	1.0
Date	24 September 2026
Audience	Product design, frontend design generation and implementation
Purpose	Premium redesign with functional parity
Canonical product name	Vaidhyaa

Brand naming note The repository and domain use Vaidhyaa, while some current screens display Vaidya. This specification uses Vaidhyaa as the canonical customer-facing name. The redesign must standardize the wordmark and must not use both spellings on the same surface.

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The section sequence follows the order a design system or frontend generator should use.

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1 How to use this specification

Treat the functional descriptions and restrictions as product requirements. Improve the presentation and interaction quality, but do not change workflow meaning, permissions, data requirements or API expectations without explicit approval.

Functional status legend

Label	Meaning for design
LIVE	Connected to production-facing API behavior and should be designed as an operable feature.
PARTIAL	Some behavior is implemented, but one or more data, persistence or permission paths remain incomplete.
DEMO	The interface exists, but the displayed data or interaction is mainly static or local.
INTERNAL	Restricted to Vaidhyaa platform operations and excluded from the clinic-facing navigation.
DESIGN REQUIREMENT	A required improvement to presentation, accessibility or responsiveness that preserves functionality.

Design authority

- The existing application behavior is authoritative for business rules unless this document marks a required correction.
- Current screenshots are references for content and workflow, not a visual template that must be preserved.
- Partial and demo features must not be represented as live production data.
- All mockups and generated examples must use synthetic clinics, patients, phone numbers, tokens and identifiers.
- Do not add clinical features, chatbot pages, exports, payment flows or analytics that are not described here.

2 Product overview

Vaidhyaa is a multi-tenant clinic operations platform with an AI voice receptionist. It brings appointment operations, doctor workflows, patient history, clinic configuration, call outcomes and approved clinic knowledge into one product.

Primary outcome

The product reduces routine reception work while keeping clinic administrators and doctors responsible for scheduling decisions, patient follow-up and clinical documentation.

Primary user groups

User	Primary job	Experience emphasis
Clinic administrator	Run clinic operations and resolve appointment actions	Operational clarity, speed and conflict prevention
Doctor	Manage the personal queue and complete consultations	Patient context, focused actions and minimal distraction
Platform administrator	Operate the multi-clinic SaaS platform	Monitoring, lifecycle controls and auditability
Support	Assist clinics within assigned permissions	Read-only visibility and explicit authorization
Patient or caller	Book, check, cancel or reschedule by voice	Clear confirmation, safety and escalation

Product qualities

- Trustworthy and appropriate for healthcare operations.
- Calm under operational pressure, with urgency shown only where action is required.
- Data-rich but easy to scan.
- Suitable for small clinics and larger multi-user teams.
- Premium and contemporary without becoming decorative or consumer-oriented.

Visual directions to avoid

- Generic admin-template layouts with interchangeable cards.
- Neon colors, heavy glassmorphism, oversized gradients or excessive blur.
- A playful chatbot visual language that reduces clinical credibility.
- Dense electronic-medical-record screens with weak hierarchy.
- Decorative charts or metrics that are not supported by real data.

3 Technical architecture

The redesigned frontend must remain compatible with the existing monorepo, API proxy and typed domain contracts.

Layer	Technology	Responsibility
Web	Next.js 15 App Router, React 19, TypeScript, Tailwind CSS	Portal UI, authentication state and same-origin API proxy
API	NestJS 10, Fastify and TypeScript	Authentication, appointments, clinic operations, calls, voice flows and jobs
Data	PostgreSQL, Drizzle package and SQL migrations	Transactional multi-tenant records
Shared	TypeScript DTOs, enums and Zod schemas	Validated contracts shared by web and API

Layer	Technology	Responsibility
Configuration	Zod environment validation	Production-safe startup and provider configuration
Testing	Vitest, Testing Library and Playwright	Unit, component and end-to-end verification
Deployment	Separate Vercel web and API projects	Independent frontend and backend deployments
Database hosting	Supabase PostgreSQL transaction pooler	Production database connectivity

Browser and API boundary

The browser calls the Next.js same-origin proxy. The proxy forwards requests to the server-only API origin. Frontend-generated code must retain this boundary.

```
Browser -> /api/backend/v1/...
Next.js server proxy -> API_BASE_URL/v1/...
NestJS API -> PostgreSQL and approved providers
```

Secrets

The browser must never receive database credentials, JWT signing secrets, provider API keys or internal service credentials.

API envelopes

The API returns a stable success or error envelope. Error designs should expose the readable message and retain the request ID when it can help support diagnose a failure.

```
Success { data: ..., meta: { request_id: 'req_xxx', debug: null } }
Error   { error: { code, message, details, request_id } }
```

Security boundaries

- Production requests use authenticated bearer tokens.
- Clinic scope is enforced by the API and must not depend on hidden fields in the UI.
- Role-based access and doctor ownership determine available actions.
- Development identity headers are not a production authentication mechanism.
- Design artifacts must not include real bearer tokens, database URLs, secrets or patient data.

4 Roles and permissions

Every screen and action must render from authenticated membership and API permission, not from a role selected visually during sign in.

Clinic administrator

- Accesses Home, Call Inbox, Appointments, Patient History, Clinic Setup, Settings and Knowledge Base.

- Confirms, cancels and reschedules appointment requests.
- Creates manual appointments and marks completed visits.
- Manages working hours, schedules, holidays, booking rules, doctors, services, fees and clinic users.
- Edits the clinic profile and configures the AI agent, languages, notification policy and knowledge.

Doctor

- Uses a focused doctor home workspace with Today, Analytics and Patient History.
- Views appointments assigned to the doctor and completes owned visits.
- Adds examination notes, diagnosis and advice when marking a visit complete.
- Edits the permitted personal schedule and views applicable booking rules.
- Must not receive clinic-wide administrative actions unless the API explicitly grants them.

Platform administrator

- Uses direct internal routes for clinic lifecycle, job health and notification operations.
- May activate or suspend clinics and inspect onboarding status.
- May apply a subscription override for the active clinic.
- Internal routes must remain separate from the clinic navigation.

Support

Support access is permission-dependent. The current frontend often groups support with administrator presentation, while several backend mutations remain restricted. Designs must include read-only, disabled and access-denied variants rather than assuming full clinic-administrator authority.

5 Information architecture

The portal has one public authentication route, role-specific clinic routes and three hidden internal platform routes.

Route	Clinic admin	Doctor	Purpose
/login	Public	Public	Sign in, clinic registration and password recovery
/	Yes	Yes	Admin home dashboard or doctor workspace
/call-inbox	Yes	No	Review AI call outcomes
/appointments	Yes	Yes	Appointment operations
/patient-history	Yes	No	Clinic-wide patient search
/clinic-setup	Yes	Yes	Role-specific clinic configuration
/settings	Yes	Yes	Role-specific preferences
/knowledge-base	Yes	No	AI knowledge authoring and review

Route	Clinic admin	Doctor	Purpose
/internal/platform/clinics	Internal	No	Clinic lifecycle and onboarding
/internal/platform/jobs	Internal	No	Queue and background-job health
/internal/platform/notifications	Internal	No	Notification monitoring

6 Global application experience

The standard portal uses a dark clinic-navigation shell and a light operational content area. The redesign should unify this shell across eligible routes and introduce complete tablet and mobile navigation.

Functional status **LIVE** The desktop shell, clinic profile and role-filtered navigation are implemented. Mobile navigation requires redesign.

Desktop shell

- Dark sidebar approximately 272 to 280 pixels wide.
- Vaidhyaa brand and AI voice receptionist descriptor.
- Clinic profile card below the brand.
- Role-filtered primary navigation.
- Signed-in user and Sign Out card near the bottom.
- Light main page with title, description and contextual actions.

Clinic profile card

Show the clinic name, immutable clinic-number badge, complete address and clinic phone. An authorized administrator or platform administrator can open the edit modal from a pencil icon.

Clinic profile modal

Editable	Not editable
Clinic name	Clinic ID
Clinic phone	Unique clinic number
Address lines	Tenant ownership
City, state, postal code and country	System identifiers

Responsive shell requirement

- Desktop uses the full sidebar.
- Tablet uses a collapsible rail or drawer.
- Mobile uses a top app bar with a navigation drawer or a role-specific bottom navigation.

- Clinic profile, user identity and Sign Out remain reachable at every width.
- The redesign must not solve mobile layout by hiding navigation.

Global system states

- Session expired, with a safe return to sign in.
- Unauthorized and forbidden access.
- Clinic suspended or inactive.
- Onboarding incomplete and agent not ready.
- Offline, stale data and retry after connection loss.
- Concurrent-update warning when another user changes the same record.

7 Authentication

Authentication combines a branded entry experience with clinic registration and password recovery.

Functional status **LIVE** Sign in, registration, email verification and password recovery are API-backed.

Sign in

- Branded healthcare visual panel and product explanation.
- Sign In and Register Clinic tabs.
- Clinic Admin and Doctor visual role cards.
- Username or email and password fields.
- Password visibility control, Forgot Password and support contact.
- Authenticated membership, not the visual role card, determines authorization.

Register clinic

Clinic information	Administrator information
Clinic name	Administrator name
Clinic phone	Email
Address, city, state and country	Mobile number
Postal code	Password

Registration verifies the email with a six-digit code and returns the newly generated immutable clinic number.

Password recovery

1. Enter the registered email address.
2. Send and verify the one-time code.
3. Create and validate a new password.
4. Confirm success and return to sign in.

Design the sending, invalid email, expired code, resend, password validation and success states.

8 Clinic administrator home

The administrator home prioritizes the current clinic day, immediate appointment decisions and a concise view of upcoming work.

Functional status **PARTIAL** Appointment actions are live. Call, callback and emergency summary metrics are not fully connected.

Page header

- Home title and operational description.
- Non-interactive AI Voice Agent status indicator.
- Add Manual Booking action.

The current header control is disabled. Redesign it as a status badge rather than an interactive switch. The real setting remains in Settings.

Operational summary

- Current user and clinic context.
- Today pending-confirmation count.
- Missed-request count.
- Total pending staff actions.
- Agent active or inactive state.

Needs your action

Show only pending appointment confirmations scheduled for the current clinic day. Each item includes patient, doctor, time, date, service, reason, status, Confirm and Cancel.

Today summary

Show current clinic-day activity such as call count and pending confirmations. Do not invent production metrics when the API does not provide them.

Next appointments

Show up to three active pending or confirmed appointments for the current day in chronological order, including patient, doctor and time.

Missed actions

Show past pending requests where the administrator did not confirm or cancel before the appointment day passed. Include patient, doctor, original date and time, service, reason, a Missed badge and No action taken.

Missed Actions is read-only. Do not display Confirm, Cancel, Reschedule or another mutation action.

9 Manual appointment workflow

The Home and Appointments entry points use the same modal and must keep identical validation and slot behavior.

Functional status **LIVE** The form, patient lookup, schedule data, slot data and appointment creation are API-backed.

Patient information

Required	Optional and validation
Patient name	Date of birth cannot be in the future
Patient phone	Age must be a number from 0 to 130
Patient age	Existing patients may be matched by phone or name

Appointment information

- Doctor and dependent active service.
- Date chosen with clinic-timezone awareness.
- Available database slot.
- Required reason for visit.
- New patient or Follow-up visit type.

Calendar and availability

- Distinguish today, selected, unavailable and holiday dates.
- Respect clinic hours, doctor hours, clinic holidays and doctor holidays.
- Show slot availability and capacity clearly.
- Handle the case where a visible slot is taken by another user before submission.

Administrator override

Override Clinic Hours is available only to an administrator. It allows manual time input and requires an override reason. It must not silently bypass capacity or double-booking protection.

Completion states

- Creating and duplicate-submit protection.
- Validation errors connected to their fields.
- Slot no longer available and capacity conflict.
- Network failure with retry that preserves entered data.
- Success feedback and refresh of affected appointment views.
- Unsaved-change confirmation before closing a populated modal.

10 Appointments workspace

This page is the primary operational view for pending, confirmed, missed, completed and requested appointment changes.

Functional status **LIVE** The appointment groups and administrator lifecycle actions are API-backed. Some doctor-facing actions require permission alignment.

Header and date control

- Today selects the current clinic date.
- A separate Clear Filter behavior should return to the default current-and-upcoming view.
- The date picker filters a selected date.
- New Appointment is visible to clinic administrators.

Pending confirmation

Show only appointments scheduled for today or a future date, grouped by doctor. Cards contain patient name and phone, visit type, source, doctor, service, reason, date, time and pending status.

Clinic administrators can Confirm or Cancel. Doctor actions must follow backend ownership and role authorization rather than presentation-only rules.

Confirmed and upcoming

Use Confirmed and Upcoming rather than Confirmed Today when the block contains future dates. Show current and future confirmed appointments, grouped by doctor.

- Mark Visited when permitted.
- Edit Date and Time when permitted.
- Cancel for clinic administrators.
- Patient and visit context appropriate to the role.

Edit date and time

1. Select a new date.
2. Load valid slots for the same doctor and service.
3. Check live capacity and schedule rules.
4. Reserve the new slot if available.
5. Release the original slot.
6. Record the appointment-change activity.

If capacity is unavailable, show a readable conflict such as This slot is full. Select another available time. Do not surface a raw database error.

Missed actions

Move past pending confirmations into a separate read-only block. Never expose Confirm or Cancel after the appointment date has passed.

Visited

Show completed visits with patient, doctor, service, date and time, reason, visit status and permitted clinical details. Marking a confirmed appointment as visited creates the patient visit record.

Reschedule and cancel requests

This administrator-only block contains pending patient requests and recent completed activity.

- Review original appointment and requested change.
- Approve the requested reschedule slot when valid.
- Choose another valid date and time, then approve.
- Reject a reschedule request and retain the original appointment.
- Approve or reject a cancellation request.
- Show before-and-after details for completed reschedules.

Appointment lifecycle

```
Pending confirmation -> Confirmed -> Visited
Pending confirmation -> Cancelled
Past pending without action -> Missed Actions
Confirmed -> Valid new slot -> Rescheduled appointment
Patient reschedule request -> Approve or Reject
Patient cancellation request -> Approve or Reject
```

11 Call inbox

The Call Inbox is a clinic-administrator review surface that combines call and action outcomes into one API-backed master-detail view.

Functional status **LIVE** The routed Call Inbox fetches the aggregated call-inbox API. It is read-only apart from filters and load retry.

Outcome filter

Use a multi-select popover with checkboxes. The trigger should show All outcomes, the selected outcome, or a selected-count summary.

- Appointment booked
- Cancelled Appointment
- Appointment Rescheduled
- General inquiry
- Callback requested
- Emergency

Master-detail layout

Call list	Details panel
Date, time and duration	Patient information
Caller phone and patient name	Call summary and outcome

Call list	Details panel
Summary and outcome badge	Required action and source status
Action-needed badge	Related appointment, callback or emergency

Data behavior

The frontend consumes an aggregated API response. It must not query physical database tables directly. The API combines appointment bookings, appointment action requests, callbacks, emergencies, inquiries and call records.

Explicit exclusions

- No Recording or Recording unavailable section.
- No Transcript or Transcript unavailable section.
- No Emergency only, Callback only, Appointment requests only or Action needed quick-filter row.
- No mutation controls unless a dedicated API action is added later.

12 Patient history

Patient history supports clinic-wide administrator search and a similar doctor-workspace experience.

Functional status **LIVE** Search and combined appointment and visit history are API-backed.

Search

Require at least one of patient phone, patient name or patient age. Validate age as an integer.

Result card

- Patient name and phone.
- Age and gender when available.
- Last visit date.
- Record count.

History modal

Show a chronological timeline combining appointments and completed visits. Include doctor, service, date and time, status, reason, examination notes, diagnosis and advice. Use Not recorded when a clinical value is absent.

Required states

- Search validation error.
- Loading and retry.
- No matching patients.
- Long-history scrolling.

- Permission-aware clinical details.
- Keyboard-accessible modal with focus restoration.

13 Clinic setup

Clinic Setup configures doctors, services, availability, holidays, booking rules and clinic users. The page changes by role.

Functional status **LIVE** The principal configuration panels and conflict previews are API-backed.

Administrator panels

1. Doctors and Services
2. Clinic Working Hours
3. Holidays
4. Doctor Schedules
5. Booking Rules
6. User Management

Doctor panels

- Own schedule editor.
- Own booking rules in read-only form unless an explicit permission is added.

Doctors and services

Doctor and service management belong in one editor. Do not create a separate Services section.

- Doctor name.
- Qualification or specialization.
- One predefined service selection.
- Consultation fee.
- Active or inactive status.
- Add, edit, activate, deactivate and permitted removal actions.

The UI displays readable names. The API and database retain the corresponding `service_key` and `service_name` values. The exact mapping appears in Appendix B.

Clinic working hours

Configure Monday through Sunday with zero, one or multiple windows per day. Users may add, remove or edit windows.

- Adding a working day or expanding availability must not create a conflict.
- Removing a working day can conflict when it strands a future active appointment.
- Shortening hours can conflict when an appointment falls outside the new window.
- Successful changes synchronize future appointment slots.

Conflict presentation

A conflict must show patient name, appointment date, time, doctor, service and reason. A raw appointment UUID is not an adequate explanation. The user resolves or reschedules affected appointments and checks again before saving.

Holidays

- Date and reason.
- Whole clinic or selected doctors.
- Active or inactive status.
- Add and remove actions.
- Conflict blocking when a future appointment is affected.

Doctor schedules

Administrators can select a doctor and edit Monday-to-Sunday windows. Doctors can edit their own permitted schedule. Doctor hours must fit inside clinic hours, and saved changes regenerate future slots.

Booking rules

- Slot duration.
- Capacity per slot.
- Booking horizon in days.
- Edit cutoff in minutes.
- Maximum shift in minutes.
- Whether the doctor may edit permitted appointment or service data.

Duration and capacity changes should preview affected appointments. When supported, the user can apply a rule from the next safe date rather than invalidate existing appointments.

User management

- List name, immutable generated username, role, doctor link and status.
- Create a clinic administrator or doctor login.
- Require a doctor account to link to an unlinked doctor.
- Enable or disable membership.
- Do not introduce a receptionist role unless the domain model is extended.

14 Settings

Settings expose role-specific agent, notification, language and subscription information.

Functional status **PARTIAL** Core settings are API-backed. Some hidden backend voice requirements are not configurable in the current UI.

Agent settings

- AI voice agent enabled or disabled.
- Booking mode: Pending confirmation or Automatic confirmation.

- Readiness explanation when onboarding or backend configuration blocks activation.

Do not reintroduce Answering Mode or Fallback Phone. Those fields were intentionally removed from this screen.

Notification settings

- Notify clinic staff when an appointment is pending.
- Explain that patients are notified after confirmation.
- Show recent events only when the API supplies them.

Do not include a Pending Notification Channel dropdown or Test Notification button.

Language settings

- Default language.
- Enabled languages.
- English remains enabled.
- Missing-template warning.
- Editable for administrators and read-only for doctors.
- Current primary options are English and Tanglish; the layout must support longer translated labels.

Subscription and limits

- Plan and subscription status.
- Trial end date.
- Voice minutes and overage.
- Concurrent-call limit.
- Recording and transcript retention.

An exact platform administrator may also see plan, status, trial and internal-notes overrides for the active clinic.

15 Knowledge base

The Knowledge Base controls what the AI receptionist may answer and separates drafts from approved runtime knowledge.

Functional status **PARTIAL** Manual Q and A, review, approval and embeddings are live. DOCX import is currently a visual prototype.

Embedding status

- Total entries.
- Ready or embedded entries.
- Pending entries.
- Failed entries.
- Coverage percentage.

Manual knowledge editor

- Question and answer.
- Category and alternative phrases.
- Source metadata and applicability.
- Save Draft and Save and Approve.
- Add a custom question.

Review queue

- Pending review and needs-update entries.
- Single and bulk selection.
- Edit, approve and disable.

Approved knowledge

Show approved question, answer preview, category, applicability, embedding status and edit or disable actions.

Categories

- Parking and facility.
- First visit and pre-visit.
- Scan and test preparation.
- Insurance.
- Reports.
- General FAQ.

Medical safety

Reject knowledge that gives diagnosis, medicine, dosage or treatment advice. Runtime answers use only approved and applicable knowledge.

DOCX import limitation

The current drag-and-drop area creates only a local processing presentation and does not upload or import the document. Designs may define intended progress and review states, but must label the integration incomplete until backend work is authorized.

16 Doctor workspace

Doctors receive a focused experience on the Home route and a separate shell from the clinic administrator.

Functional status **PARTIAL** Queue retrieval, history and visit completion are live. Several queue actions and analytics remain local or static.

Doctor navigation

- Today.
- Analytics.
- Patient History.
- Logout.

The current sections are local tabs rather than routes. A premium redesign may recommend URL-based subsections for deep linking, but must preserve access behavior unless routing work is approved.

Today queue

- Selected date and queue totals.
- Waiting, In Progress, Visited and Skipped groups.
- Patient, time, service, reason, visit type and appointment state.
- Previous-visit and completed-visit details.

Consultation workspace

- Patient identity and reason for visit.
- Previous visit context.
- Examination notes.
- Diagnosis.
- Advice.
- Start Consultation, Skip, Save Draft and Mark Visited actions.

Mark Visited persists the completed visit and clinical information. Start Consultation, Skip and Save Draft are currently mainly client-side states.

Analytics

The current interface contains KPI cards, daily, weekly and monthly filters, trend charts, top conditions, visit types, age distribution and a monthly table. The underlying values are demo data. Design the production-ready system without presenting the sample values as live analytics.

Patient history

Use the patient search and combined appointment and clinical timeline described in Section 12, presented within the doctor workspace.

Permission rule

The doctor sees only ownership-scoped actions authorized by the API. Do not display Confirm, Cancel or Edit Date and Time merely because those controls exist on a reusable appointment card.

17 AI voice receptionist

The backend supports a text-first, voice-ready clinic receptionist that resolves clinic identity from the called number and performs safe operational tasks.

Functional status **PARTIAL** Core conversation tools and voice lifecycle exist, while deployment readiness depends on clinic and provider configuration.

Supported capabilities

- Resolve the clinic from the called phone number.
- Provide clinic hours, location, fees and general information.
- List services and doctors.
- Recommend a service from the stated reason without diagnosing.
- Check live slots and create an appointment after explicit confirmation.
- Check appointment status.
- Cancel an appointment after explicit confirmation.
- Submit a reschedule request.
- Search approved clinic knowledge.
- Record callback requests and emergency incidents.
- Store call outcome and summary.

Booking conversation

1. Understand the reason for the visit.
2. Select or recommend a service.
3. Select a doctor.
4. Select a date and find an available slot.
5. Collect patient information.
6. Repeat the details and ask for explicit confirmation.
7. Create the appointment and provide the final status.

Safety behavior

- Never diagnose a patient.
- Never recommend medicine, dosage or treatment.
- During an emergency, advise calling emergency services, including 108 where appropriate, or proceeding to an emergency facility.
- Create an emergency incident and stop the normal booking flow.
- Escalate mental-health distress or requests for a human through the configured safe path.

18 Internal platform operations

Platform screens support Vaidhyaa staff and should use a distinct internal operations shell rather than the clinic sidebar.

Functional status **INTERNAL** These routes are restricted and absent from clinic navigation.

Clinic operations

- Clinic list with name, onboarding state and status.
- Onboarding checklist modal.
- Activate or suspend a clinic.
- Checklist areas include clinic details, administrator, hours, doctors, services, mappings, schedules, rules, knowledge, telephony and test conversation.

Job health

- Queue mode and Redis state.
- Pending and failed job counts.
- Last runs for slot generation, hold expiry, recording cleanup and transcript cleanup.
- Recent run table and retry-on-load behavior.

Notification operations

- Filter by status, event type and clinic.
- Show date, clinic, event, channel, recipient, status, attempts and error.
- Retry eligible failures and cancel eligible non-terminal events.
- Display a masked payload when the API provides it.

Internal design guidance

Use denser tables and operational filters than the clinic portal, but retain the same component tokens, accessibility rules and status taxonomy. Make internal scope visually clear.

19 End-to-end workflows

These workflows define the sequence and decision points the redesign must preserve.

Manual booking

1. Open the shared manual-appointment modal.
2. Enter or find the patient.
3. Select doctor and active service.
4. Select a valid date and live slot.
5. Enter reason and visit type.
6. Validate availability again on submit.
7. Create the appointment and refresh affected views.

Reschedule confirmed appointment

1. Open Edit Date and Time for an authorized appointment.
2. Select date and load real available slots.

3. Choose an available slot.
4. Recheck capacity on save.
5. Move the appointment, release the old slot and record the activity.

Change clinic or doctor availability

1. Edit working days or time windows.
2. Compare the proposed availability with future active appointments.
3. Allow additions or expansions without false conflicts.
4. Show detailed conflicts for removals or contractions.
5. Resolve affected appointments and save again.
6. Regenerate future slots from the saved configuration.

Complete consultation

1. Open an owned confirmed appointment.
2. Review patient and previous-visit context.
3. Record examination notes, diagnosis and advice.
4. Mark the appointment visited.
5. Persist the patient visit and move the record to Visited.

Knowledge approval

1. Create or edit a knowledge entry.
2. Save it for review or approve it if authorized.
3. Reject unsafe medical content.
4. Generate or refresh the embedding.
5. Make only approved and applicable entries available at runtime.

20 Domain model and status taxonomy

The frontend should use these entity groups and semantic statuses consistently across screens.

Core entities

Area	Key records
Clinic	Profile, unique number, address, settings, telephony, onboarding and subscription
Access	User, clinic membership, role, doctor link, status and session
Clinical setup	Doctor, service, mapping, fee, schedule and booking rules
Availability	Clinic hours, doctor hours, holidays, blocked periods, slots and holds

Area	Key records
Appointments	Patient, doctor, service, time, source, status, request, event and visit
Voice operations	Call, conversation, outcome, callback, emergency and notification
Knowledge	Question, answer, category, approval, applicability, language and embedding

Semantic status system

Status	Meaning	Recommended color
Pending	Awaiting an authorized decision	Amber
Confirmed	Appointment accepted and active	Emerald
Visited	Consultation completed	Blue
Missed	Past pending request without action	Rose
Cancelled	Appointment no longer active	Red
Rescheduled	Time or date changed	Violet
Callback required	Staff follow-up required	Orange
Emergency	Urgent escalation	Strong red
Inactive	Disabled or unavailable	Slate
Draft	Saved but not approved	Neutral gray
Needs review	Human review required	Amber

Always pair color with a readable status label and, where helpful, an icon. Never rely on color alone.

21 Premium design system

The redesign should establish one coherent visual language across authentication, clinic operations, doctor workflows and internal platform tools.

Functional status **DESIGN REQUIREMENT** The current project has useful colors and layout conventions but no complete reusable component system.

Visual personality

- Calm clinical confidence with clear operational priority.
- Dark navigation, warm white surfaces and soft slate backgrounds.
- Evergreen or teal as the principal brand accent.
- Semantic colors reserved for status and risk.
- Subtle elevation and generous spacing rather than decorative effects.

Recommended palette

Purpose	Hex	Use
Sidebar and ink	#0B1220	Navigation and highest-emphasis text
Primary text	#0F172A	Headings and body emphasis
Secondary text	#64748B	Metadata and supporting copy
Page background	#F6F8FB	Primary application canvas
Card background	#FFFFFF	Content surfaces
Border	#E2E8F0	Dividers and field outlines
Primary brand	#0F766E	Primary action and active navigation
Brand accent	#14B8A6	Selected highlights and progress
Success	#15803D	Confirmed and successful
Warning	#D97706	Pending and review
Danger	#DC2626	Destructive and emergency
Information	#2563EB	Visited and informational
Reschedule	#7C3AED	Appointment change activity

Typography

- Use one modern interface font such as Inter, Geist or Manrope.
- Page title: 30 to 36 pixels, bold.
- Section title: 20 to 24 pixels, semibold or bold.
- Card title: 16 to 18 pixels, semibold.
- Body: 14 to 16 pixels.
- Metadata and labels: 12 to 13 pixels with adequate contrast.
- A restrained serif may appear only in selected brand or authentication moments.

Spacing and shape

- Use an 8-pixel spacing grid.
- Field radius: 10 to 12 pixels.
- Card radius: approximately 16 pixels.
- Large panels and modals: 20 to 24 pixels.
- Use thin consistent borders and subtle shadows only where hierarchy requires them.
- Do not round every element or place every sentence inside a card.

Component library

- Application shell, sidebar, mobile navigation and page header.
- Clinic profile card and profile editor.

- Button, icon button, input, textarea, phone input, select, multi-select, checkbox, radio and switch.
- Date picker, time picker and slot picker.
- Card, statistic card, appointment card, call card and patient card.
- Status badge, filter bar, tabs, table and pagination.
- Loading skeleton, empty state, error state and toast.
- Modal, drawer, confirmation dialog, conflict panel and activity timeline.
- Permission-denied and read-only states.

Each component requires default, hover, focus, active, disabled, loading and error variants, plus mobile behavior.

Icons and motion

- Use one outline icon library such as Lucide or Phosphor.
- Do not mix emoji, Unicode symbols and unrelated SVG styles.
- Use subtle 150 to 220 millisecond transitions for state changes.
- Respect reduced-motion preferences.
- Avoid animated decoration that competes with clinical actions.

22 Responsive and accessible behavior

The generated experience must be fully usable on desktop, tablet and mobile and should target WCAG 2.1 AA.

Responsive behavior

Viewport	Required behavior
Desktop	Full sidebar, multi-column operational cards, master-detail inbox and data tables
Tablet	Collapsible navigation, adaptive one- or two-column content and accessible filters
Mobile	Real mobile navigation, single-column cards, full-screen long forms and sticky actions where useful

Accessibility

- Visible keyboard focus and logical tab order.
- Semantic buttons instead of clickable div elements.
- Accessible labels for icon-only controls.
- Errors connected to the relevant form field.
- Proper table headers and screen-reader announcements.
- Focus trap, Escape behavior and focus restoration for dialogs.
- At least 44-pixel touch targets.
- Sufficient contrast and non-color status indicators.
- Reduced-motion support.
- Layouts resilient to long names, addresses and translated content.

Modal standard

- Render through a portal and lock background scrolling.
- Provide a visible title and accessible description.
- Trap focus and return focus to the trigger.
- Close with Escape when safe and always provide an explicit Close or Cancel action.
- Confirm before discarding populated forms.
- Keep the primary action reachable.
- Use a full-screen pattern or bottom sheet for long mobile forms.

Loading, empty and error states

- Use contextual skeletons for page and list loading.
- Provide an empty-state explanation and relevant next action.
- Distinguish validation, permission, conflict and network failures.
- Offer Retry and preserve entered data where possible.
- Display the API request ID when it can help support.
- Prevent duplicate submission and show Saving or Creating states.
- Use privacy-conscious toast content without sensitive patient details.

23 Implementation constraints and maturity

The redesign must protect business rules and clearly separate live behavior from partial or demo functionality.

Rules that must not change

1. Do not change role permissions for visual convenience.
2. Do not allow clinic ID or unique clinic number editing.
3. Do not show Confirm or Cancel in Missed Actions.
4. Do not show old requests in current pending or confirmed sections.
5. Do not reintroduce Call Inbox recordings, transcripts or removed quick filters.
6. Do not create a separate Services section in Clinic Setup.
7. Do not reintroduce Answering Mode or Fallback Phone in Agent Settings.
8. Do not reintroduce notification-channel selection or Test Notification.
9. Do not present the Home agent status as an operable switch.
10. Do not invent analytics, metrics or clinical data.
11. Do not bypass slot capacity, working hours or double-booking protection.
12. Do not show a raw appointment ID as the only conflict explanation.
13. Do not expose server secrets or real patient data.
14. Do not allow the AI receptionist to provide medical advice.
15. Do not create false conflicts when availability is expanded.
16. Do not silently invalidate appointments when availability is contracted.

17. Keep Home and Appointments manual booking behavior identical.
18. Use clinic timezone for today, missed and upcoming calculations.

Implementation maturity

Status	Areas
Substantially live	Authentication, clinic profile, appointment lifecycle, manual booking, slots, schedules, conflicts, holidays, patient history, user management, languages, call inbox and manual knowledge review
Partial	Agent readiness, Home call totals, doctor queue persistence, selected doctor actions, notification detail and some platform operations
Demo or local	Doctor analytics, doctor duty presentation, Start Consultation, Skip, Save Draft and some patient-detail fallbacks
Visual prototype	DOCX knowledge import
Internal only	Clinic operations, job health, notification monitoring and subscription override

Known permission and readiness gaps

- Some doctor-facing appointment controls can appear before the backend grants the corresponding lifecycle action. The design must derive action visibility from real permission.
- Support presentation and backend mutation permissions are not fully aligned.
- New-clinic voice readiness depends on backend fields that are no longer exposed in the settings screen.
- Normal platform-user sign-in and clinic selection require further product alignment.

24 Design delivery requirements

A design-generation system should return a developer-ready product system, not only a set of attractive dashboard images.

1. A unified product design system with tokens for color, typography, spacing, radius, elevation and motion.
2. Desktop, tablet and mobile application shells.
3. Clinic-administrator, doctor and internal-platform screen families.
4. High-fidelity layouts for every screen in this specification.
5. Reusable component variants and interaction documentation.
6. Loading, empty, error, permission, conflict, disabled, saving and success states.
7. Appointment and call status taxonomy used consistently across pages.
8. Accessible keyboard, focus and screen-reader behavior.
9. Responsive specifications and mobile navigation.

10. Developer-ready component breakdown and data dependencies.
11. If code is generated, Next.js 15, React 19, TypeScript and Tailwind-compatible components.
12. No business-logic, role, route or API-contract changes without explicit approval.

Acceptance checklist

- Every route has a documented desktop, tablet and mobile composition.
- Every operational block has loading, empty, error and populated states.
- Every destructive action has confirmation and safe recovery language.
- Every status uses text and not color alone.
- Long names, addresses and multilingual content do not break layout.
- All mock data is visibly synthetic.
- No removed fields or controls return in generated designs.
- Partial features are labeled and not represented as live data.

A Routes and source reference

These paths help a developer connect the design specification to the current implementation.

Area	Current implementation path
Routes	apps/web/src/app
Navigation policy	apps/web/src/lib/navigation.ts
Application shell	apps/web/src/components/layout/AppShell.tsx
Admin home	apps/web/src/components/pages/HomePageContent.tsx
Appointments	apps/web/src/components/pages/ AppointmentsPageContent.tsx
Manual booking	apps/web/src/components/pages/appointments/ ManualAppointmentModal.tsx
Call inbox	apps/web/src/components/pages/call-inbox/ CallInboxPageContent.tsx
Clinic setup	apps/web/src/components/pages/clinic-setup
Settings	apps/web/src/components/pages/settings
Doctor workspace	apps/web/src/components/doctor
Knowledge base	apps/web/src/components/pages/knowledge-base
API modules	apps/api/src/modules
Shared contracts	packages/shared/src
Database package	packages/db

B Predefined service mapping

The dropdown displays the service name while the system stores the corresponding service key and service name.

service_key	service_name
orthopaedic_consultation	Orthopaedic Consultation
paediatric_consultation	Paediatric Consultation
general_consultation	General Consultation
dermatology_consultation	Dermatology Consultation
ent_consultation	ENT Consultation
dental_consultation	Dental Consultation
cardiology_consultation	Cardiology Consultation
ophthalmology_consultation	Ophthalmology Consultation
gynaecology_consultation	Gynaecology Consultation
neurology_consultation	Neurology Consultation
gastroenterology_consultation	Gastroenterology Consultation

C API response format

Frontend-generated components should consume the standard envelopes and preserve readable errors and request IDs.

Successful response

```
{
  "data": {},
  "meta": {
    "request_id": "req_xxx",
    "debug": null
  }
}
```

Error response

```
{
  "error": {
    "code": "ERROR_CODE",
    "message": "Readable message",
    "details": {},
    "request_id": "req_xxx"
  }
}
```

D Copy-ready prompt for the design system

Paste this instruction together with the rest of the specification into the frontend design-generation application.

Redesign Vaidhyaa as a premium healthcare SaaS application using this specification as the functional source of truth. Preserve every workflow, role restriction, business rule, state transition, field and API expectation. Improve the visual system, usability, responsiveness, accessibility, navigation, component consistency, feedback states and information hierarchy. Use a calm, trustworthy clinical style built from deep navy, evergreen or teal, warm white, soft slate and restrained semantic colors. Avoid generic dashboard styling, excessive gradients, glassmorphism, decorative clutter, fake data and unapproved functionality. Create a unified design system and developer-ready desktop, tablet and mobile screens for clinic administrators, doctors and internal platform administrators. Include all loading, empty, error, conflict, permission, saving, success and disabled states. Do not change business logic or expose secrets. Where this specification marks a feature as partial, demo or internal, represent that status accurately and do not imply a completed production integration.

End of specification